Crime Analytics – Big Data Project Report

Crime in Karnataka 2025 – End-to-End Big Data Pipeline (HDFS/PySpark + MySQL/SQLite + Streamlit)

1. Objectives

- Distributed ETL: HDFS -> PySpark -> Spark SQL -> MySQL/SQLite -> Streamlit
- Extract/clean/normalize NCRB-style district-level crime data (Karnataka 2025)
- Compute KPIs/rates, trends, domain-wise analysis for data-driven decisions

2. Data Sources & Validation

Source: Crime in Karnataka 2025 (Govt. of Karnataka Police Department)
Extraction: pdfplumber -> 26 CSVs in crime_karnataka_2025_csv/
Validation: 1,521 trend rows, 39 districts, 2023–2025; 13 normalized tables
Reference artifacts: CSV zip, DB (crime_analytics.db), schema.sql, data_dictionary.csv

3. Architecture & Methodology

Pipeline: HDFS (raw/processed) -> PySpark (clean/typecast/dedup) -> Spark SQL (aggregations) -> SQLite/MySQL (results) -> Streamlit (UI)
Components: scripts/hdfs_upload.sh, pyspark_crime_analytics.py, load_to_sqlite.py, streamlit_app_sqlite.py, streamlit_app.py, schema.sql

4. Analytics, Insights & Interpretation

- Fact records: 1521 (domain x district x year)
- Coverage: 39 districts (2023–2025)
- Period: 2023–2025
- Total cognizable cases (trend): 2023:428468, 2024:396662, 2025:404538
- Top 5 by total crime rate (2025): Shimoga (602.5); Bengaluru District (520.0); Kolar (487.1); Tumakuru (441.5); Mysuru District (397.7)
- Top 5 domains (2025): sll (123472); violent (43254); crime_against_women (40904); cyber_crime (28792); economic_offences (14594)
- Avg property recovery % (2025): 49.04 (enforcement effectiveness)
- Avg IPC/BNS recidivism % (2025): 3.97 (repeat offenders)
- 5Vs of Big Data: Volume (structured fact), Variety (multiple crime domains), Velocity (batch ETL), Veracity (cleaned/typed/deduped), Value (actionable KPIs)
- Insights: Inter-district variation indicates localized drivers; domain mix informs resource allocation; recovery % and recidivism % measure outcomes, not just incidents

5. Conclusion & Future Scope

End-to-end big data analytics pipeline implemented, validated, and documented. Interactive Streamlit dashboard supports exploratory analysis. Future: productionize with live HDFS/Spark/MySQL, forecasting, hotspot mapping, station-level drilldowns.